



Final Year Project Proposal

Digital FIR

A Blockchain Based Secure First Information Report Management System

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1. Introduction

The First Information Report, commonly known as an FIR, is the document that formally opens the criminal justice process in Pakistan. It records a reported offence, identifies the parties involved and initiates investigation. Because so many stakeholders touch an FIR over its lifetime, ranging from the complainant to the investigating officer, the Station House Officer and supervisory staff, maintaining a consistent and trustworthy record of what happened and when is essential.

Police departments across Pakistan have already invested in digitizing parts of this process. Punjab, Sindh and Khyber Pakhtunkhwa each operate their own complaint and record management platforms, described in Section 3. These systems demonstrate that digital complaint handling and case record management are achievable at scale. What they generally rely on, however, is a conventional centralized database model, where the integrity of a record depends on trusted administrators and application level audit logs rather than on an independently verifiable mechanism.

This project proposes a Digital FIR Management System that keeps the familiar workflow of complaint submission, FIR registration, investigation and closure but adds a permissioned blockchain layer, built on Hyperledger Fabric, purely for integrity, traceability and tamper evident auditing. Sensitive information about complainants, suspects and case narratives never leaves protected off chain storage. Only cryptographic hashes, transaction identifiers, workflow events and evidence metadata are written to the blockchain. The result is a system that can prove, independently of any single administrator, that a record has or has not been altered since it was created.

This proposal is written as an academic prototype. It uses synthetic data throughout, is not connected to any live police infrastructure, NADRA, or criminal record system and does not claim any legal standing. Its purpose is to investigate, build and evaluate the technical feasibility of the hybrid architecture described above.

2. Executive Summary

The proposed Digital FIR Management System is a secure web-based platform that digitizes the lifecycle of First Information Reports for a conceptual Pakistan Police environment. It combines a conventional relational database with a permissioned Hyperledger Fabric blockchain network. Sensitive FIR information stays in protected off chain storage, while important transactions, record hashes, workflow events and integrity metadata are recorded on the blockchain.

The purpose of the project is not simply to place FIR data on a blockchain. It is to investigate how a blockchain layer can add integrity, traceability, accountability and tamper evident auditing on top of an FIR workflow that already resembles the systems in operational use today. The platform provides separate functionality for citizens, police officers, investigating officers, Station House Officers, supervisory officers and system administrators and supports the full journey of a complaint from submission through investigation to closure.

Project Definition

A secure web based Digital FIR management platform that combines Python with FastAPI, PostgreSQL, role based access control and Hyperledger Fabric to provide permissioned, tamper evident auditability and cryptographic integrity verification for FIR workflows and their associated digital documents, while keeping sensitive information in protected off chain storage.

3. Background and Existing Systems in Pakistan

Before proposing a new system, it is important to establish what already exists. Pakistan's provincial police organizations have implemented several digital initiatives and this proposal should not be read as claiming that digital FIR or police management systems are absent in Pakistan. Instead, the sections below summarize what is publicly known about current systems, so that the actual research gap can be stated precisely in Section 5.

3.1 Punjab Police

Punjab Police operates a Complaint Management System (CMS) for lodging and tracking complaints and a Police Station Record Management System (PSRMS) that is integrated with FIR related records. The IGP Complaint Center, reachable at the short code 1787, allows citizens to submit complaints through multiple channels and track their progress. Punjab Police also runs the Police Khidmat Markaz network, which provides services such as FIR copies and crime reports through citizen facilitation centers.

3.2 Sindh Police

Sindh Police has implemented a Police Station Record Management System that is reported to be operational across 643 police stations, electronically recording FIRs and their corresponding register entries. In 2024, Sindh Police also launched a Complaint Management System intended to digitize complaint handling and to provide a foundation for further digital FIR processes.

3.3 Khyber Pakhtunkhwa Police

Khyber Pakhtunkhwa Police operates its own Police Station Record Management System together with several digital policing services. Its IGP Helpline accepts complaints through multiple channels and generates unique, trackable complaint identifiers, with timeline based notifications for delayed or pending complaints. The KP Police website also offers an online FIR registration option.

3.4 Comparative Analysis of Existing Systems

Table 1 compares the publicly documented capabilities of these three provincial systems against the system proposed in this document. The comparison is not intended to suggest that existing systems are insecure or ineffective; rather, it shows where the proposed academic prototype adds a capability that is not identified as a core, publicly documented feature of current systems, specifically the permissioned blockchain integrity layer.

Feature	Punjab	Sindh	KP	Proposed
Digital complaint submission	✓	✓	✓	✓
Digital FIR and case record management	✓	✓	✓	✓
Complaint or case tracking	✓	✓	✓	✓
Role based access control	✓	✓	✓	✓
Officer or investigation assignment	✓	✓	✓	✓
Digital evidence metadata	Limited	Limited	Limited	✓

Feature	Punjab	Sindh	KP	Proposed
Cryptographic document hashing	Not documented	Not documented	Not documented	✓
Permissioned blockchain ledger	✗	✗	✗	✓
Blockchain based transaction history	✗	✗	✗	✓
Tamper evident audit trail	Conventional logs	Conventional logs	Conventional logs	Blockchain and logs
Off chain storage of sensitive data	✓	✓	✓	✓
Independent document integrity verification	Not documented	Not documented	Not documented	✓ (SHA-256)

Table 1: Feature comparison between existing provincial systems and the proposed Digital FIR system

The comparison supports a narrower and more defensible research gap than assuming that no digitization exists: existing Pakistani police systems already provide digital complaint, FIR and record management capabilities, while this project investigates whether a permissioned blockchain layer can strengthen integrity verification, transaction traceability, evidence metadata integrity and tamper evident auditing on top of that existing style of workflow.

4. Problem Statement

Existing police information systems have increasingly adopted computerized complaint, FIR and record management mechanisms. Conventional centralized systems of this kind generally rely on trusted database administrators and application-level audit mechanisms to protect the record. For sensitive FIR workflows, however, it is important to be able to demonstrate, independently of any single administrator, reliable evidence of who created a record, who modified it, when an operation occurred, which officer performed it, what workflow state changed, whether an associated document was altered and whether the audit record itself can be verified.

This project addresses that gap by proposing a permissioned blockchain backed Digital FIR system that combines conventional database storage with cryptographically verifiable blockchain transactions, so that the integrity of critical operations can be checked independently of the application layer that recorded them.

5. Research Gap and Novelty

As shown in Section 3, existing Pakistani police systems already demonstrate digital complaints, FIR record management, complaint tracking, police record digitization, supervisory monitoring and public service facilitation. The proposed system does not attempt to duplicate this. Its contribution is the investigation of permissioned blockchain technology for FIR auditability, specifically blockchain-based transaction history, cryptographic document integrity, blockchain backed evidence metadata, dual layer authorization spanning both the application and the chaincode and a hybrid on chain and off chain architecture.

The novelty of the project, therefore, is not digital FIR registration by itself. It is the integration of an FIR workflow, a permissioned blockchain, cryptographic integrity checks, role-based authorization and auditable evidence metadata into a single, coherent architecture, evaluated through functional, security and performance testing.

6. Aim and Objectives

Aim: To design and develop a secure, blockchain enabled Digital FIR Management System that provides controlled access, tamper evident auditability, case traceability and cryptographic integrity verification for FIR related records in a conceptual Pakistan Police environment.

Objectives:

ID	Objective
O1	Develop a digital interface for authorized personnel to create FIR records.
O2	Implement secure authentication and identity management for all system users.
O3	Restrict system functionality according to user role through server-side authorization.
O4	Integrate a permissioned Hyperledger Fabric network with the web application.

ID	Objective
O5	Develop chaincode to manage important FIR transactions on the blockchain.
O6	Use cryptographic hashing to detect unauthorized modification of FIR records and documents.
O7	Allow authorized users to track an FIR from registration through investigation and closure.
O8	Register evidence metadata and document hashes so later modification can be detected.
O9	Maintain a reliable, chronological history of important FIR operations.
O10	Provide dashboards and reports for authorized supervisory users.
O11	Notify relevant users automatically when important FIR events occur.
O12	Protect confidential information through encryption, authentication, authorization and controlled data access.

7. Stakeholder Analysis

Table 3 identifies the people and groups who have an interest in the system, what that interest is and how strongly they interact with it during the academic prototype phase.

Stakeholder	Interest	Interaction
Citizens and complainants	Submit and track complaints	High
Police officers	Register and process FIRs	High
Investigating officers	Investigate assigned FIRs	High
Station House Officer	Approve, assign and supervise FIRs	High
Supervisory officers	Monitor multiple stations and review cases	High
Police department	Administrative oversight	High
System administrators	Technical management of the platform	High
Investigation and audit authority	Audit and independently verify records	Medium to High

Development team	System development and maintenance	High
University and project supervisor	Academic evaluation	Medium
Government and police IT authorities	Potential future institutional adopter	Future
Legal and policy authorities	Compliance and legal evaluation	Future

Table 3: Stakeholder analysis

The primary stakeholders, whose needs directly shape the requirements in this document, are citizens, police officers, investigating officers, Station House Officers, supervisory officers and system administrators. Secondary stakeholders, whose interest is real but indirect during the prototype phase, include the police IT department, audit authorities, the university and project supervisor and, further in the future, legal and policy authorities and potential government technology partners.

8. Scope of the Project

8.1 Revised Scope Statement

The scope of the proposed project is to design and develop a secure web based Digital FIR Management System for a conceptual Pakistan Police environment that manages the complete digital FIR lifecycle, from complaint submission and FIR registration through verification, approval, investigation, evidence and document registration, supervisory review and closure. The system implements role-based access control, secure REST APIs, encrypted off chain storage, cryptographic document hashing, permissioned Hyperledger Fabric transactions, blockchain backed audit trails, evidence integrity verification, notifications, dashboards and reporting. Throughout, the platform demonstrates how a permissioned blockchain can serve as a tamper evident integrity and audit layer while sensitive FIR information remains in protected off chain storage.

8.2 In Scope

The scope covers eleven functional modules, summarized in Table 4.

Module	Coverage
Authentication and identity	Registration, login, password security, session tokens, role and account management
Citizen portal	Complaint submission, complaint tracking, notifications, access to authorized FIR information
FIR management	FIR creation, validation, approval or rejection, numbering, status tracking, search and filtering
Police station management	Station information, officers, Station House Officer functions, assignments, station level dashboard
Investigation management	Investigating officer assignment, investigation events, progress updates, investigation documents, closure requests
Evidence management	Evidence registration, metadata, document upload, SHA-256 hashing, integrity verification and status
Blockchain	Permissioned Fabric network, chaincode, transaction and hash recording, transaction identifiers, history retrieval
Audit	Chronological audit trail, blockchain transaction history, verification, integrity checks
Notifications	Alerts for submission, approval, assignment, updates, evidence addition, closure and integrity failures
Analytics	FIR statistics, status distribution, station and investigation statistics, processing time, evidence statistics
Administration	Users, roles, permissions, stations, system configuration, technical monitoring

Table 4: Functional modules included in the project scope

8.3 Out of Scope

The following are explicitly excluded from the academic prototype:

- Integration with live Pakistan Police systems or real police databases
- Real CNIC verification through NADRA, or real criminal history data
- Real world legal authorization or operational deployment of FIRs
- AI based profiling, facial recognition, or automated determination of guilt
- Replacement of any existing government system

All testing uses synthetic or demonstration data only.

9. Proposed Solution

9.1 Design Principle

The solution is a web based Digital FIR Management System integrated with a permissioned Hyperledger Fabric network. Information is deliberately split into two categories: sensitive data that must remain confidential and integrity metadata that benefits from independent, tamper evident verification. Hyperledger Fabric was chosen over a public blockchain because it supports identifiable organizational identities, controlled permissions through Membership Service Providers, endorsement policies and private channels, all of which are better suited to sensitive police records than a fully public and transparent ledger.

9.2 On Chain Information

The blockchain stores only: the FIR unique identifier, transaction identifier, FIR record hash, document or evidence hash, operation type, current FIR status, officer or organization identifier, timestamp, workflow state, approval information, audit events and evidence integrity metadata.

9.3 Off Chain Information

Sensitive data is stored outside the blockchain, in PostgreSQL and protected file storage: complainant personal information, CNIC information, contact information, the FIR narrative, suspect information, witness information, investigation notes, uploaded documents, evidence files, images, PDFs and other confidential information. If a document is later modified, its newly calculated hash will differ from the hash recorded on the blockchain, allowing the system to detect the change.

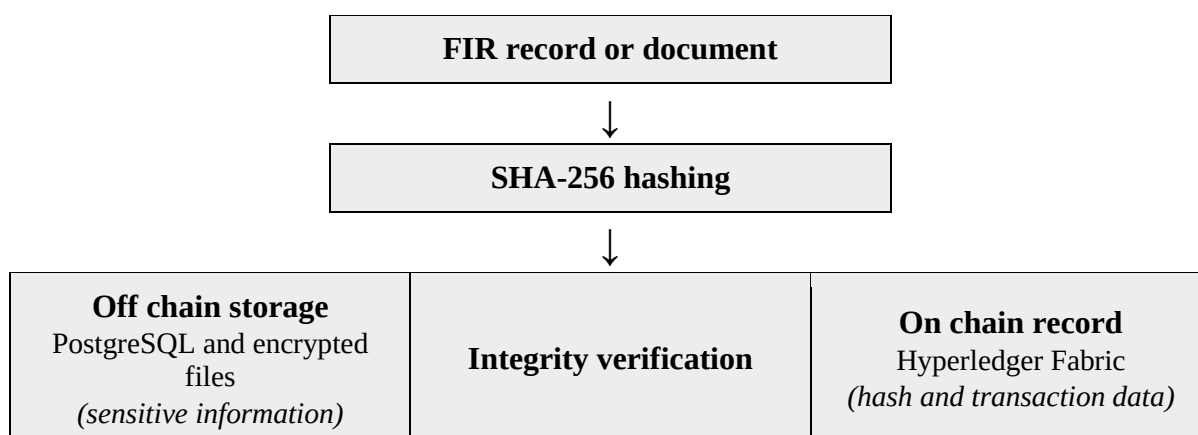


Figure 1: Integrity verification flow

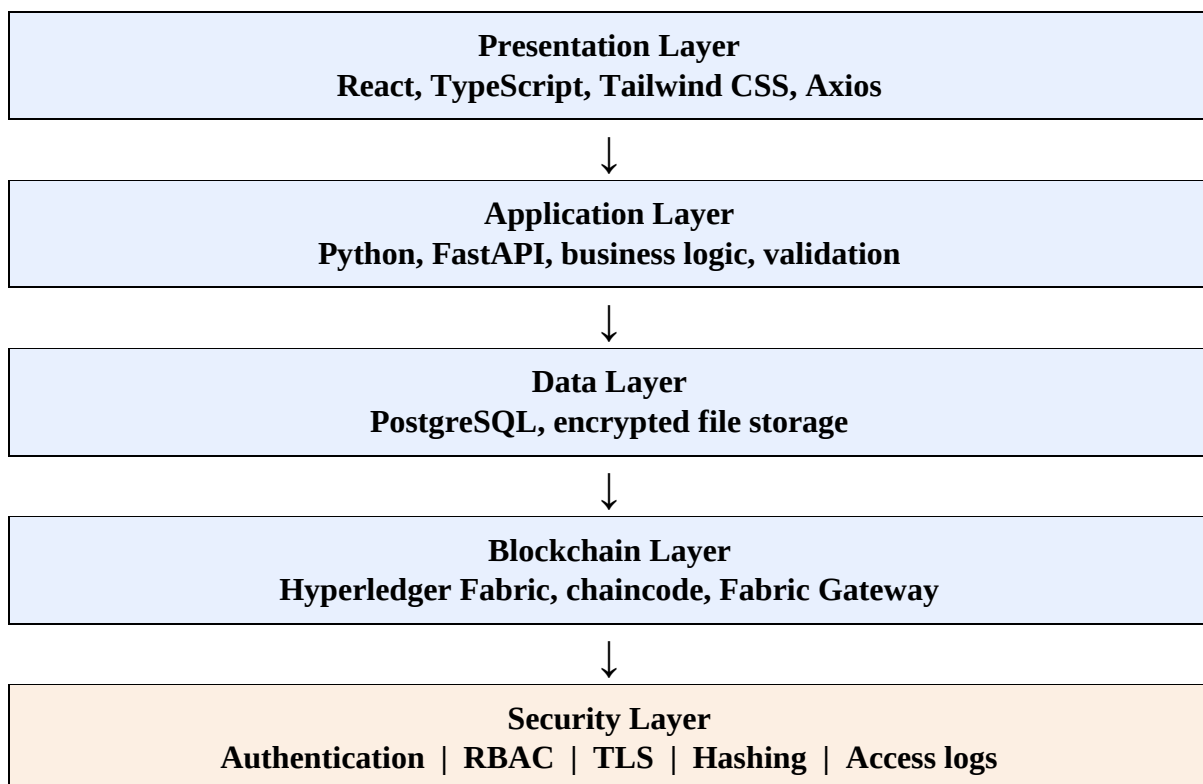
10. System Architecture

10.1 Architecture Layers

The system is organized into five layers.

- Presentation layer: React, TypeScript, Tailwind CSS, Axios and React Router, delivering role specific interfaces to citizens and police personnel.
- Application layer: Python and Fast API, implementing authentication, authorization, FIR business logic, validation, evidence management and notifications.
- Data layer: PostgreSQL for structured records, together with encrypted file storage for documents and evidence.
- Blockchain layer: Hyperledger Fabric, including peers, an ordering service, a certificate authority, Membership Service Providers, channels and chain code.
- Security layer: Authentication, role-based access control, TLS, password hashing, input validation and access logging, applied consistently across every other layer.

10.2 High Level Architecture Diagram



11. FIR Lifecycle

Figure 2 shows how an FIR moves through the system. Every transition marked with a solid arrow generates an auditable blockchain transaction, so a chronological history of the case can always be reconstructed independently of the application database.

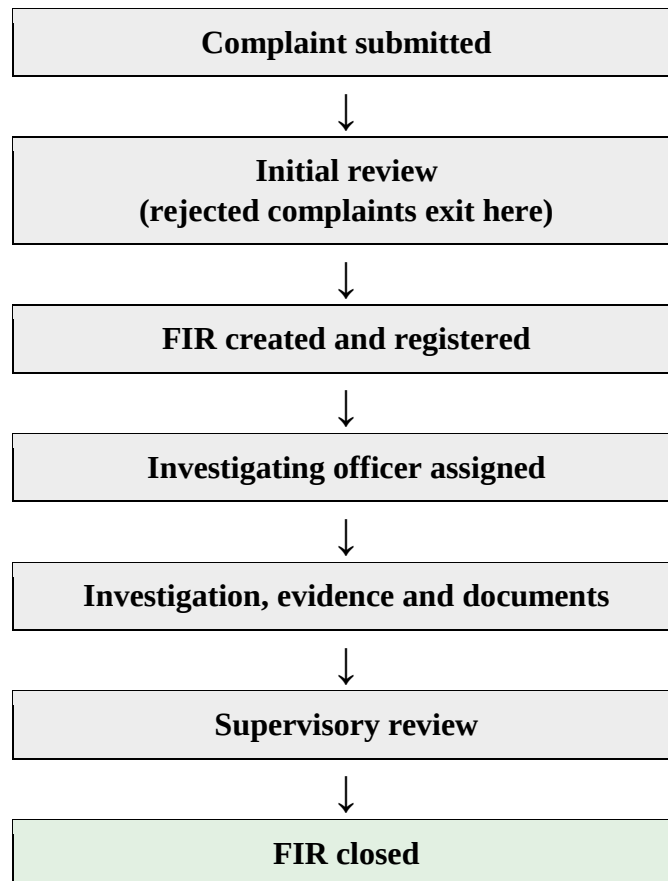


Figure 2: FIR lifecycle from submission to closure

12. Functional Requirements

Authentication

ID	Requirement
FR1	The system shall authenticate registered users using secure credentials.
FR2	The system shall issue a secure authenticated session token after successful authentication.
FR3	The system shall reject invalid credentials.
FR4	The system shall support account activation and deactivation.

Authorization

ID	Requirement
FR5	The system shall implement role based access control.
FR6	The system shall verify authorization on the backend before executing any protected operation.
FR7	Users shall only access FIRs permitted by their role and organizational scope.

Complaint Management

ID	Requirement
FR8	Citizens shall be able to submit complaints.
FR9	The system shall generate a unique complaint identifier.
FR10	Citizens shall be able to track complaint status.

FIR Management

ID	Requirement
FR11	Authorized police personnel shall create FIR records.
FR12	The system shall generate a unique FIR identifier or number.
FR13	The system shall validate mandatory FIR information.
FR14	Authorized officers shall review FIR submissions.
FR15	Authorized officers shall approve or reject applicable FIR submissions.
FR16	The system shall maintain FIR status.

FIR Workflow

ID	Requirement
FR17	The system shall support defined FIR states, including Draft, Submitted, Under Review, Registered, Assigned, Investigation, Pending Review and Closed.
FR18	Invalid status transitions shall be rejected.

Officer Assignment

ID	Requirement
FR19	Authorized supervisory personnel shall assign an investigating officer.
FR20	The system shall maintain assignment history.

Investigation

ID	Requirement
FR21	Investigating officers shall record investigation events.
FR22	Investigation events shall include a timestamp and the responsible user.
FR23	Authorized officers shall update investigation status.

Evidence

ID	Requirement
FR24	Authorized users shall register evidence metadata.
FR25	The system shall generate a unique evidence identifier.
FR26	The system shall calculate SHA-256 hashes for uploaded documents.
FR27	Evidence metadata and hash references shall be associated with the relevant FIR.

Document Management

ID	Requirement
FR28	Authorized users shall upload permitted documents.
FR29	The system shall enforce file type and file size restrictions.
FR30	The system shall calculate document hashes.
FR31	The system shall verify document integrity against the recorded hash.

Blockchain

ID	Requirement
FR32	Important FIR operations shall generate blockchain transactions.
FR33	The system shall record relevant hashes on Hyperledger Fabric.
FR34	The system shall store the blockchain transaction identifier.

ID	Requirement
FR35	Authorized users shall retrieve blockchain transaction history.

Audit

ID	Requirement
FR36	The system shall maintain a chronological audit trail.
FR37	Audit events shall identify user, role, action, timestamp and FIR.
FR38	Blockchain transaction references shall be associated with applicable audit events.

Search

ID	Requirement
FR39	Authorized users shall search FIRs.
FR40	The system shall support filtering by FIR number, date, status, station, category and assigned officer.

Notifications

ID	Requirement
FR41	The system shall generate notifications for significant workflow events.

Reporting

ID	Requirement
FR42	Authorized users shall generate FIR and investigation reports.

Dashboard

ID	Requirement
FR43	The system shall provide role specific dashboards.

Closure

ID	Requirement
FR44	Authorized personnel shall request FIR closure.
FR45	Authorized supervisory personnel shall approve closure.

Table 5: Functional requirements

13. Nonfunctional Requirements

Table 6 lists the nonfunctional requirements together with a measurable or observable target for each, so that they can be verified during testing rather than assessed only by opinion.

ID	Requirement	Target
NFR1	Security	All protected functions require authentication and authorization
NFR2	Password security	Passwords stored using Argon2 or bcrypt
NFR3	API security	Unauthorized API requests are rejected
NFR4	Confidentiality	Sensitive traffic protected using TLS
NFR5	Integrity	Documents verified using SHA-256
NFR6	Auditability	Security sensitive actions are logged
NFR7	Availability	System remains operational during normal testing
NFR8	Performance	Common API operations meet defined response targets under a test workload
NFR9	Scalability	Architecture supports additional organizational units
NFR10	Maintainability	Modular frontend, backend and blockchain architecture
NFR11	Usability	Interfaces are understandable to the intended users
NFR12	Reliability	Invalid transactions do not corrupt application state
NFR13	Compatibility	Modern browsers are supported
NFR14	Portability	Deployable on a Linux or Ubuntu environment
NFR15	Privacy	Sensitive personal data is kept off chain
NFR16	Recoverability	Database and configuration support backup and recovery
NFR17	Traceability	Major workflow operations are traceable
NFR18	Data validation	User input and uploaded files are validated

ID	Requirement	Target
NFR19	Least privilege	Users receive only the permissions their role requires
NFR20	Logging	Security and application events are logged

Table 6: Nonfunctional requirements

14. Security Requirements and Testing Strategy

Because this project handles simulated police and citizen data, security is treated as a first class requirement rather than an afterthought.

14.1 Security Requirements

- Authentication: secure password hashing, JWT based sessions, token expiration, secure logout and account activation or deactivation.
- Authorization: role-based access control enforced on the backend, resource level authorization, least privilege access and organization or station level restrictions.
- API security: protection against SQL injection, cross site scripting, cross site request forgery where applicable, broken access control, malicious input, parameter tampering, unauthorized endpoint access, brute force login attempts and excessive request abuse.
- File security: file extension and MIME type validation, file size limits, filename sanitization, storage isolation, hash generation and access control on stored files.
- Database security: parameterized queries or an ORM, restricted database permissions, credentials kept in environment variables, backup procedures and no plaintext passwords.
- Blockchain security: X.509 identities, the Fabric certificate authority, Membership Service Providers, endorsement policies, channel level access policies, chaincode authorization and transaction validation.
- Data privacy: CNIC numbers, addresses, phone numbers, personal information, investigation notes and evidence files remain off chain at all times.

14.2 Security Testing Strategy

- Authentication testing: valid login, invalid login, empty credentials, repeated failed login and expired, invalid, or modified JWT tokens.

- Authorization testing: a citizen attempting to access a police endpoint, a police officer attempting to access an administrator endpoint, an investigating officer attempting to access another officer's restricted case and supervisory access limited to permitted organizational records. The expected result in every case is an unauthorized request returning an HTTP 401 or 403 response, the operation being rejected and the event being logged.
- API security testing: SQL injection attempts, cross site scripting payloads, invalid JSON, missing parameters, oversized requests, parameter manipulation and access attempts against unauthorized endpoints.
- File security testing: executable file upload, invalid extensions, oversized files, corrupted documents, duplicate documents and modified documents.
- Blockchain testing: valid transactions, unauthorized transactions, invalid status transitions, invalid identities, chaincode validation, transaction history retrieval and hash verification.
- Integrity demonstration: the team will intentionally modify a test document after its original hash has been recorded on the blockchain. The expected result is that the original hash no longer matches the current hash and the system reports integrity verification as failed. This demonstration is one of the clearest proofs of the project's value.

14.3 Security Testing Tools

Tool	Purpose
Postman	API testing
OWASP ZAP	Web application security testing
Burp Suite Community	HTTP and API security testing
pytest	Backend unit and integration testing
pytest-cov	Code coverage measurement
Playwright or Selenium	User interface testing
Bandit	Python static security analysis
Trivy	Container vulnerability scanning

Tool	Purpose
OpenSSL	Certificate and TLS testing
Git and GitHub	Version control and code review
Docker	Isolated deployment and testing

Table 7: Security and quality testing tools

15. Technology Stack

Layer	Technologies
Frontend	React, TypeScript, HTML5, CSS3, Tailwind CSS, Axios, React Router, Recharts
Backend	Python 3.11 or later, Fast API, Unicorn, Pedantic, SQL Alchemy, Alembic, JWT authentication, Argon2 or crypt password hashing
Database and storage	PostgreSQL for structured data, protected file storage for documents and evidence
Blockchain	Hyperledger Fabric, including Fabric peers, an ordered, a certificate authority, Membership Service Providers, channels, the Fabric Gateway and chaincode written in Node.js and TypeScript
Security and DevOps	SHA-256, JWT, HTTPS and TLS, Docker, Docker Compose, Git, GitHub, Postman

Table 8: Technology stack by layer

Why Fast API. Fast API was selected for the backend because it offers high-performance REST APIs, automatic Open API documentation, Pedantic based validation, a mature Python ecosystem, clean modular architecture, straightforward integration with PostgreSQL and good support for asynchronous operations.

Why Hyperledger Fabric. Fabric was selected over a public blockchain because police records require identifiable organizational identities, controlled permissions, endorsement policies and private channels, none of which a fully public and transparent ledger provides. The application backend is implemented in Python, while Fabric chain code is implemented in Node.js and TypeScript, which is Fabric's most mature

chaincode language; the project does not force a single language across every component where it is not the natural fit.

16. Research Questions

RQ1. How can permissioned blockchain technology improve the integrity and auditability of digital FIR records?

RQ2. How can sensitive FIR information be protected using a hybrid on chain and off chain architecture?

RQ3. How can role-based access control be integrated with permissioned blockchain identities in a digital police workflow?

RQ4. How effectively can SHA-256 hashing detect unauthorized modification of FIR related documents?

RQ5. What performance overhead is introduced by blockchain transaction processing compared with conventional database operations?

RQ6. Can blockchain backed audit trails provide stronger traceability for critical FIR workflow events?

17. Development Methodology

The project follows an Agile and Scrum inspired iterative methodology, divided into the twelve sprints shown in Table 9.

Table 9: Sprint plan

Sprint	Focus Area
Sprint 1	Requirements and research
Sprint 2	System architecture and database design
Sprint 3	Frontend foundation
Sprint 4	Backend APIs using Python and Fast API
Sprint 5	Authentication and role-based access control
Sprint 6	FIR management
Sprint 7	Hyperledger Fabric network setup
Sprint 8	Chaincode development

Sprint	Focus Area
Sprint 9	Blockchain integration
Sprint 10	Evidence and hash verification
Sprint 11	Testing and security
Sprint 12	Deployment, documentation and presentation

18. Proposed Project Timeline

Table 10: Proposed project timeline

Phase	Duration
Research, requirements and literature review	2 weeks
System design, database design and architecture	2 weeks
Backend foundation and authentication	2 to 3 weeks
Frontend foundation and core FIR modules	3 weeks
Hyperledger Fabric network setup and chaincode	3 to 4 weeks
Blockchain integration and evidence integrity	2 weeks
Testing, security checks and refinement	2 weeks
Documentation, deployment packaging and presentation	2 weeks

19. Team and Responsibilities

Member	SAP ID	Primary Responsibility
Sharaiz Ahmed	57288	System architecture and blockchain integration
Abdul Moiz	54482	Backend development, database and APIs
Basit Ali	54596	Frontend development, UI and UX and testing

Table 11: Team members and primary responsibility

Sharaiz Ahmed, System Architecture and Blockchain Lead. Overall architecture, requirements coordination, the Hyperledger Fabric network, Fabric organizations, the

certificate authority and Membership Service Providers, chaincode coordination, blockchain integration, hash and integrity architecture, security architecture, system integration, GitHub repository management and final deployment coordination.

Abdul Moiz, Backend and Database Lead. Python and FastAPI development, REST APIs, PostgreSQL, SQLAlchemy, Alembic migrations, authentication, role based access control, FIR APIs, investigation APIs, evidence APIs, notification APIs and backend testing.

Basit Ali, Frontend and Testing Lead. The React application, TypeScript, UI and UX design, the login interface, dashboards, FIR forms, the investigation interface, the evidence interface, the audit interface, charts, responsive design, UI testing and usability testing.

Shared Responsibilities. All three members participate in requirement analysis, research, documentation, GitHub commits, code review, integration, security testing, system testing, the final report, the presentation and viva preparation.

20. Minimum Viable Product

If time is constrained, the following are treated as mandatory:

- Secure login and role based access control
- Citizen complaint submission
- FIR creation, approval and status tracking
- Officer assignment and basic investigation updates
- Evidence registration with SHA-256 hashing
- Hyperledger Fabric integration and recording of key transactions
- Blockchain history retrieval and document integrity verification
- A basic audit trail

Secondary, in order of priority if time allows: dashboards, notifications, reports and analytics and an advanced audit interface.

Optional and explicitly not allowed to consume time needed for the core: email or SMS alerts, multi factor authentication, a mobile application and advanced analytics.

21. Proposed Data Model

FIR.: FIR identifier, FIR number, complaint identifier, police station identifier, crime category, date and time reported, incident date, incident location, description, complainant identifier, assigned investigating officer, status, created at, updated at, record hash and blockchain transaction identifier.

Complainant: Complainant identifier, name, CNIC, phone, address and email. Sensitive fields remain off chain.

Investigation: Investigation identifier, FIR identifier, officer identifier, action, description, timestamp, document reference and hash.

Evidence: Evidence identifier, FIR identifier, evidence type, description, collected by, collection time, file reference, file hash and blockchain transaction identifier.

Audit Event: Event identifier, FIR identifier, user identifier, role, action, timestamp, transaction identifier, previous state, new state and record hash.

22. Success Criteria and Evaluation Metrics

The project will be considered successful if:

- Authorized users can authenticate and roles are correctly enforced.
- Authorized users can create and progress FIR records through the defined workflow states.
- Investigating officers can be assigned and investigation events recorded.
- Evidence metadata can be registered and document hashes generated.
- Important operations successfully produce blockchain transactions.
- Blockchain history can be retrieved and document integrity can be verified against recorded hashes.
- Unauthorized actions are rejected by the application, the chaincode, or both.
- The complete prototype can be demonstrated end to end using synthetic data.

Metric	Evaluation Method
Functional correctness	Percentage of passed test cases
Authentication	Unauthorized login attempts rejected
Authorization	Role based access control test cases

Metric	Evaluation Method
Integrity	Hash verification accuracy
Blockchain	Successful valid transactions
Security	Vulnerability test results
Performance	API and transaction response time
Reliability	Failure and recovery tests
Usability	User questionnaire
Maintainability	Code and documentation review
Auditability	Ability to reconstruct FIR history

Table 12: Evaluation metrics

23. Limitations

- The system is an academic prototype, not a production police system.
- Only synthetic or demonstration data is used; no real citizen or police records are involved.
- There is no integration with NADRA, live police databases, or operational government systems.
- The blockchain network is small and controlled, built for demonstration purposes.
- Performance and scale testing is limited to the infrastructure available to the student team.
- The prototype does not provide legally binding digital FIR functionality.

24. Future Enhancements

Beyond the current scope, possible future directions include NADRA or government identity integration, inter station or provincial data sharing under proper authority, a mobile application with SMS and email notifications, stronger digital signature and PKI support, interfaces for courts or prosecutors and multilingual Urdu and English interfaces with more advanced analytics.

25. Expected Benefits of the Prototype

The project is expected to demonstrate improved record integrity through cryptographic verification, stronger accountability through role-based controls and blockchain audit trails, clearer visibility into the FIR lifecycle and a practical example of applying permissioned blockchain to a public sector style workflow while keeping sensitive data off chain.

26. Conclusion

This proposal outlines a BSCS Final Year Project to design and develop a Digital FIR Management System that combines a conventional secure database with a permissioned Hyperledger Fabric blockchain. Existing Pakistani police organizations, including Punjab Police, Sindh Police and Khyber Pakhtunkhwa Police, already operate computerized complaint and record management systems and this proposal does not attempt to duplicate that work. Instead, it introduces a research-oriented architecture that combines Digital FIR management, Python and Fast API, PostgreSQL, role-based access control, cryptographic hashing and Hyperledger Fabric, in order to investigate whether a permissioned blockchain layer can improve the integrity, traceability and auditability of critical FIR transactions and digital evidence, while keeping sensitive information off chain.

Sensitive information will remain off chain at all times; the blockchain will provide a tamper evident record of important transactions, status changes and document hashes. The system will be implemented with a React frontend and a Python with Fast API backend, supported by PostgreSQL and a small Fabric test network and evaluated through functional, integration, performance, usability and security testing using synthetic data. Success will be measured by the project's ability to demonstrate a secure role-based FIR workflow with blockchain backed auditability and cryptographic integrity verification.

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